MATTHEW FROSST

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PROFESSIONAL SUMMARY

Quantitative researcher and PhD candidate in computational astrophysics with experience in statistical modelling, large-scale data analysis, and high-performance computing (HPC). Known for leading international research teams, building reproducible HPC pipelines, and delivering actionable insights from some of the worlds largest datasets. Now seeking to use advanced computational, analytical, and leadership skills in a consulting environment – particularly in strategy, data, or AI-implementation domains.

EXPERIENCE

PhD Candidate (Computational Astrophysics) — Perth, Australia
University of Western Australia, International Centre for Radio Astronomy Research — *2022 – Nov 2025*

- Developed custom simulation tools using Python, C++, and R to model galaxy formation and evolution, validated against large observational datasets.
- Managed projects involving 150+ million CPU-hours; optimised workflows to reduce runtime by 5-10%.
- Applied Bayesian inference, regression, and dimensionality reduction to extract insights from 100Tb+ of structured and unstructured data.
- Led international collaborations, coordinating multi-disciplinary teams to ensure efficient project delivery.

European Southern Observatory – Scholarship Recipient — Munich, Germany
European Southern Observatory — *March 2025 - June 2025*

- Recipient of prestigious Science Support Discretionary Fund Scholarship to undertake research with European partners.
- Developed advanced model-to-observation comparison tools by applying knowledge of HPC and forward modelling to introduce observational errors into simulations.
- Built and stress-tested predictive models using simulated "mock" observations and compared them with real observational data.
- Advised multi-national teams on data integrity and backend system reliability, ensuring reproducibility and compliance.

PhD Candidates' Representative — Perth, Australia
International Centre for Radio Astronomy Research, University of Western Australia — *2024 - 2025*

- Drove reform of milestone assessments, improving program retention and completion rates.
- Advocated for 60+ students in senior staff strategic meetings, securing $45,000 publication fund.
- Demonstrated ability to manage large teams and deliver change in complex institutional environments.

Computational-Theory Group Chair — Perth, Australia
International Centre for Radio Astronomy Research, University of Western Australia — *2022 - 2024*

- Organised 15+ member research unit focused on theoretical astrophysics, numerical simulations, and high-performance computing.
- Designed and led technical and non-technical events, from workshops and speaker presentations to community events (i.e., monthly BBQ's and an annual half-marathon charity run).

Consultant Data Scientist — *Kingston, Canada*
Queen's University — *2017 - 2019*

- Delivered interactive analytics dashboards and performance tracking systems using Python and SQL.
- Presented analytic insights to senior faculty, leading to measurable improvements in student retention and instructional quality.

SKILLS AND INTERESTS

Programming	Expert in Python (numpy, pandas, scikit-learn, matplotlib, pydantic-ai), R, C++, SQL, Fortran
Infrastructure	Expert in GNU/Linux, Git, HPC systems, Slurm, Bash scripting
Methods	Expert in machine learning, Bayesian inference, regression, MCMC, simulation design
Communication	Excellent technical writing, stakeholder briefings / presentations, public science outreach skills
Personal Interests	Ironman triathlon, chess, astronomy education, rock climbing

SELECTED AWARDS & GRANTS

2025	**Science Support Discretionary Fund**, European Southern Observatory	*$ 8,500*
2024	**Graduate Research School Travel Award**, University of Western Australia	*$ 1,850*
2024	**Student Travel Assistance Scheme**, Astronomical Society of Australia (ASA)	*$ 1,500*
2022	**Research Training Program Stipend**, Australian Government	*$225,000*

SERVICE, OUTREACH, AND VOLUNTEERING

ICRAR Student Day Conference hair — Perth, Australia
University of Western Australia — *2024*

- Chair of the local organising committee of a day-long graduate student focused conference for 50+ attendees.

ASA Annual Science Meeting Committee — Perth, Australia
University of Western Australia — *2024*

- Helped to organise the Astronomical Society of Australia Annual Science Meeting, hosting over 500+ national and international scientists.
- Oversaw the finance budget. Event was significantly under budget.

Diversity, Equity, and Inclusion (DEI) Committee Student Representative — Perth, Australia
University of Western Australia — *2023*

- Participated as student representative on the ICRAR DEI committee.
- Liaison to ICRAR Visiting Fellow for Senior Women+ in Astronomy.
- Organised sessions to help the professional development of students and early-career researchers.
- Championed the creation of the *Renu Sharma Scholarship in Astronomy and Astrophysics*, providing $10,000 for women, Aboriginal and Torres Strait Islanders studying astronomy and astrophysics at the University of Western Australia.

EDUCATION AND PUBLICATIONS

PhD (Computational Astrophysics) — Perth, Australia
University of Western Australia, International Centre for Radio Astronomy Research — *2022 – Nov 2025*

Authored 7 peer-reviewed publications on quantitative modelling, statistical inference, and large-scale data analysis. Example articles include:

Origins and lifetimes of secular and tidal bars in simulated disc galaxies — *August 2025*
Frosst, M., Obreschkow, D., Ludlow, A., and Fraser-McKelvie, A.

The complex relationship between black hole feedback, star formation, and stellar bars in TNG50 — *March 2025*
Frosst, M., Obreschkow, D., Ludlow, A., Bottrell, C., and Genel, S.

MSc (Physics, Engineering Physics & Astronomy) — Kingston, Canada
Queen's University — *2019 – 2022*

Authored 1 peer-reviewed publication comparing simulated and observed galaxy properties:

Diversity of spiral galaxies explained — *August 2022*
Frosst, M., Courteau, S., Arora, N., Stone, C., Maccio, A. V., and Blank, M.

BSc Honours (Physics, Engineering Physics & Astronomy) — Kingston, Canada
Queen's University — *2015 – 2019*

TEACHING

Teaching Assistant (PHYS 104/106) — *Kingston, Canada*
Queen's University — *2019-2022*

- Organised and taught cohorts of 30+ students during 8-month long lab-based introductory physics classes.
- Lectures were delivered twice a week, and were accompanied by a period in which students undertook both procedural based and student design based experimental activities under my supervision.

Undergraduate Physics Tutor — *Kingston, Canada*
Queen's University — *2021-2022*

- Personally tutored small groups of undergraduate students in basic and advanced physics.
- Students found great success in the tutored courses, on averaged showing a 18% increase in final exam grade semester-over-semester.